



**TAKORADI
TECHNICAL
UNIVERSITY**

FACULTY OF APPLIED SCIENCES

DEPARTMENT OF COMPUTER SCIENCE

END OF FIRST SEMESTER EXAMINATIONS 2023/2024 ACADEMIC YEAR

CANDIDATE'S INDEX NUMBER

CLASS & YEAR

4Y BTECH IT - (SE Option)

DURATION

3 HOURS

**COURSE
TITLE**

**DATA STRUCTURES AND
ALGORITHMS**

**COURSE
CODE**

BITS 335

INSTRUCTIONS - ANSWER ALL QUESTIONS.

SECTION A (1 Hour 30 minutes) – (60 Marks)- Answer all questions

Question 1

- (a) Explain data structure. **[4 marks]**
- (b) Describe Linear and Non-linear data structures and state one example of each. **[4 marks]**
- (c) Describe the following data structures:
- a. Stack.
 - b. Queue
 - c. Linked list
 - d. Array
 - e. Graph
 - f. Binary tree **[12 marks]**

[Total marks 20]

Question 2

(a) Explain the following data structure operations:

a. Insertion

b. Deletion

[4 marks]

(b) Describe an algorithm. Why is there a need to write an algorithm before coding a program? [4 marks]

(c) Describe top-down and bottom-up approaches for solving problems. [4 marks]

(d) Suppose a salesman earns a basic wage of GH¢ 5000 in a week. He also earns a commission of 25% of the value of all the goods he sells. If he sells more than GH¢ 5000 of goods in any week, he receives a bonus of GH¢ 250. Write the algorithm. [4 marks]

(e) Describe the following Searching and Sorting algorithms:

a. Binary search

b. Quick sort

[4 marks]

[Total marks 20]

Question 3

(a) Explain time and space complexities of algorithm. [4 marks]

(b) Develop an algorithm to add two numeric values from a user. Note that your algorithm should make sure that the inputs are only numeric types before addition is done. [4 marks]

(c) Develop an algorithm to print string "ALGORITHM" in the reverse order. [4 marks]

(d) Develop an algorithm to compare four numeric values to determine the following:

a. The smallest number

b. The largest number

[4 marks]

(e) Create an implementation of a sequential search algorithm in any programming language. [4 marks]

[Total marks 20]

SECTION B (1 Hour 30 minutes) – (40 Marks)

1. Create a computer program in any programming language of your choice to perform the following operations on a singly linked list that stores floating point values:
 - i. Traverses and prints the elements of the linked list backwards; that is starting from the last element of the list to the first. **[8 marks]**
 - ii. Inserts an element at the last position of the linked list. **[8 marks]**
 - iii. Deletes the first element of the linked list. **[8 marks]**
 - iv. Requests the user to enter a number, and searches the first half of the linked list for the number using binary search algorithm. **[8 marks]**
 - v. Sorts the elements of the linked list in descending order using the Bubble sort algorithm. **[8 marks]**

[Total marks 40]